MSG No. 176.148 P_c6_3/m [Type IV, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `6'/m..`

No.	position	mapping
1	$\left[0,\, 0,\, \frac{3}{4}\right]$	`[1,3,5,8,10,12,14,16,18,19,21,23]`
2	$\left[0,\, 0,\, \frac{1}{4}\right]$	`[2,4,6,7,9,11,13,15,17,20,22,24]`

Table 2: Wyckoff site: `2b`, site symmetry: `6'/m'..`

No.	position	mapping
1	$[0,\, 0,\, 0]$	`[1,3,5,7,9,11,14,16,18,20,22,24]`
2	$\left[0,\, 0,\, \frac{1}{2}\right]$	`[2,4,6,8,10,12,13,15,17,19,21,23]`

Table 3: Wyckoff site: `4c`, site symmetry: `-6..`

No.	position	mapping
1	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{3}{4}\right]$	`[1,3,5,8,10,12]`
2	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{1}{4}\right]$	`[2,4,6,7,9,11]`
3	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{1}{4}\right]$	`[13,15,17,20,22,24]`
4	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{3}{4}\right]$	`[14,16,18,19,21,23]`

Table 4: Wyckoff site: `4d`, site symmetry: `-6'..`

No.	position	mapping
1	$\left[\frac{1}{3},\, \frac{2}{3},\, 0\right]$	`[1,3,5,20,22,24]`
2	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{1}{2}\right]$	`[2,4,6,19,21,23]`
3	$\left[\frac{2}{3},\, \frac{1}{3},\, 0\right]$	`[7,9,11,14,16,18]`
4	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{1}{2}\right]$	`[8,10,12,13,15,17]`

Table 5: Wyckoff site: `4e`, site symmetry: `6'..`

No.	position	mapping
1	$[0,\, 0,\, z]$	`[1,3,5,14,16,18]`
2	$\left[0,\, 0,\, z+\frac{1}{2}\right]$	`[2,4,6,13,15,17]`
3	$[0,\, 0,\, -z]$	`[7,9,11,20,22,24]`
4	$\left[0,\, 0,\, \frac{1}{2}-z\right]$	`[8,10,12,19,21,23]`

Table 6: Wyckoff site: `6f`, site symmetry: `2'/m..`

No.	position	mapping
1	$\left[\frac{1}{2},\, 0,\, \frac{3}{4}\right]$	`[1,10,16,19]`
2	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{4}\right]$	`[2,11,17,20]`
3	$\left[0,\, \frac{1}{2},\, \frac{3}{4}\right]$	`[3,12,18,21]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[4,7,13,22]
5	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	[5,8,14,23]
6	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[6,9,15,24]

Table 7: Wyckoff site: 6g, site symmetry: 2'/m'..

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,7,16,22]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[2,8,17,23]
3	$\left[0, \frac{1}{2}, 0\right]$	[3,9,18,24]
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[4,10,13,19]
5	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[5,11,14,20]
6	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[6,12,15,21]

Table 8: Wyckoff site: 8h, site symmetry: 3..

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	[1,3,5]
2	$\left[\frac{2}{3}, \frac{1}{3}, z+\frac{1}{2}\right]$	[2,4,6]
3	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	[7,9,11]
4	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}-z\right]$	[8,10,12]
5	$\left[\frac{1}{3}, \frac{2}{3}, z+\frac{1}{2}\right]$	[13,15,17]
6	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	[14,16,18]
7	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}-z\right]$	[19,21,23]
8	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	[20,22,24]

Table 9: Wyckoff site: 12i, site symmetry: 2'..

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	[1,16]
2	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	[2,17]
3	$\left[0, \frac{1}{2}, z\right]$	[3,18]
4	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	[4,13]
5	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[5,14]
6	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	[6,15]
7	$\left[\frac{1}{2}, 0, -z\right]$	[7,22]
8	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-z\right]$	[8,23]
9	$\left[0, \frac{1}{2}, -z\right]$	[9,24]
10	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	[10,19]
11	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[11,20]
12	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	[12,21]

Table 10: Wyckoff site: `12j`, site symmetry: `m..`

No.	position	mapping
1	$\left[x,\, y,\, \frac{3}{4}\right]$	`[1,10]`
2	$\left[x-y,\, x,\, \frac{1}{4}\right]$	`[2,11]`
3	$\left[-y,\, x-y,\, \frac{3}{4}\right]$	`[3,12]`
4	$\left[-x,\, -y,\, \frac{1}{4}\right]$	`[4,7]`
5	$\left[-x+y,\, -x,\, \frac{3}{4}\right]$	`[5,8]`
6	$\left[y,\, -x+y,\, \frac{1}{4}\right]$	`[6,9]`
7	$\left[x,\, y,\, \frac{1}{4}\right]$	`[13,22]`
8	$\left[x-y,\, x,\, \frac{3}{4}\right]$	`[14,23]`
9	$\left[-y,\, x-y,\, \frac{1}{4}\right]$	`[15,24]`
10	$\left[-x,\, -y,\, \frac{3}{4}\right]$	`[16,19]`
11	$\left[-x+y,\, -x,\, \frac{1}{4}\right]$	`[17,20]`
12	$\left[y,\, -x+y,\, \frac{3}{4}\right]$	`[18,21]`

Table 11: Wyckoff site: `12k`, site symmetry: `m'..`

No.	position	mapping
1	$[x,\, y,\, 0]$	`[1,22]`
2	$\left[x-y,\, x,\, \frac{1}{2}\right]$	`[2,23]`
3	$[-y,\, x-y,\, 0]$	`[3,24]`
4	$\left[-x,\, -y,\, \frac{1}{2}\right]$	`[4,19]`
5	$[-x+y,\, -x,\, 0]$	`[5,20]`
6	$\left[y,\, -x+y,\, \frac{1}{2}\right]$	`[6,21]`
7	$[-x,\, -y,\, 0]$	`[7,16]`
8	$\left[-x+y,\, -x,\, \frac{1}{2}\right]$	`[8,17]`
9	$[y,\, -x+y,\, 0]$	`[9,18]`
10	$\left[x,\, y,\, \frac{1}{2}\right]$	`[10,13]`
11	$[x-y,\, x,\, 0]$	`[11,14]`
12	$\left[-y,\, x-y,\, \frac{1}{2}\right]$	`[12,15]`

Table 12: Wyckoff site: `24l`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$\left[x-y,\, x,\, z+\frac{1}{2}\right]$	`[2]`
3	$[-y,\, x-y,\, z]$	`[3]`
4	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	`[4]`
5	$[-x+y,\, -x,\, z]$	`[5]`
6	$\left[y,\, -x+y,\, z+\frac{1}{2}\right]$	`[6]`
7	$[-x,\, -y,\, -z]$	`[7]`
8	$\left[-x+y,\, -x,\, \frac{1}{2}-z\right]$	`[8]`
9	$[y,\, -x+y,\, -z]$	`[9]`
10	$\left[x,\, y,\, \frac{1}{2}-z\right]$	`[10]`
11	$[x-y,\, x,\, -z]$	`[11]`
12	$\left[-y,\, x-y,\, \frac{1}{2}-z\right]$	`[12]`

continued ...

Table 12

No.	position	mapping
13	$\left[x,\, y,\, z+\frac{1}{2}\right]$	[13]
14	$[x-y,\, x,\, z]$	[14]
15	$\left[-y,\, x-y,\, z+\frac{1}{2}\right]$	[15]
16	$[-x,\, -y,\, z]$	[16]
17	$\left[-x+y,\, -x,\, z+\frac{1}{2}\right]$	[17]
18	$[y,\, -x+y,\, z]$	[18]
19	$\left[-x,\, -y,\, \frac{1}{2}-z\right]$	[19]
20	$[-x+y,\, -x,\, -z]$	[20]
21	$\left[y,\, -x+y,\, \frac{1}{2}-z\right]$	[21]
22	$[x,\, y,\, -z]$	[22]
23	$\left[x-y,\, x,\, \frac{1}{2}-z\right]$	[23]
24	$[-y,\, x-y,\, -z]$	[24]